ROVERS

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What is a rover ?

A rover is a robotic vehicle which is constructed to travel over unpredictable terrain to carry out specific tasks, explore an area and gather scientific data. Rovers are sent to places too dangerous, far, or impractical for humans to visit, and they send back images, samples, and scientific readings.
A rover can operate for many months or even years collecting geological data and transmitting it to Earth without ever having to eat, breathe or make its way back home.



Fig: 1: Rover

Sensors used on a Rover

Navigation and movement :

- Cameras- They act as the rovers eyes also used for navigation, avoiding obstacles, and capturing images for analysis. High definition and stereo cameras are used
- LiDAR and Sonar- Rangefinders bounce light or sound waves off nearby objects to help the rover avoid crashes. Used for terrain mapping and obstacle detection.

- IMU (Inertial Measurement Unit)- Combines an accelerometer and gyroscope to track orientation, tilt, and balance. Gyroscopes and accelerometers track the rovers tilt, speed, and orientation.
- Wheel encoders- Track how far the rover has moved and help detect wheel slip.

Weather:

- Temperature and humidity- Monitor environmental conditions.
- Spectrometers- Used on planetary rovers to analyze the chemical composition of soil and rock samples.
- Radiation Detectors- Measure high-energy radiation from space and the sun to check surface safety.
- Thermal Sensors- Infrared radiometers and thermocouples track heat distribution on rocks and soil.

Scientific:

- Spectrometers- Used on planetary rovers to analyze the chemical composition of soil and rock samples.
- Current and voltage sensors- Keep track of the rover's power system and battery health.
- Ground-Penetrating Radar- Sends radar waves underground to map subsurface layers and search for buried ice or water.
- Neutron Detectors- Measure bouncing neutrons to detect hydrogen beneath the surface, pointing to hidden water

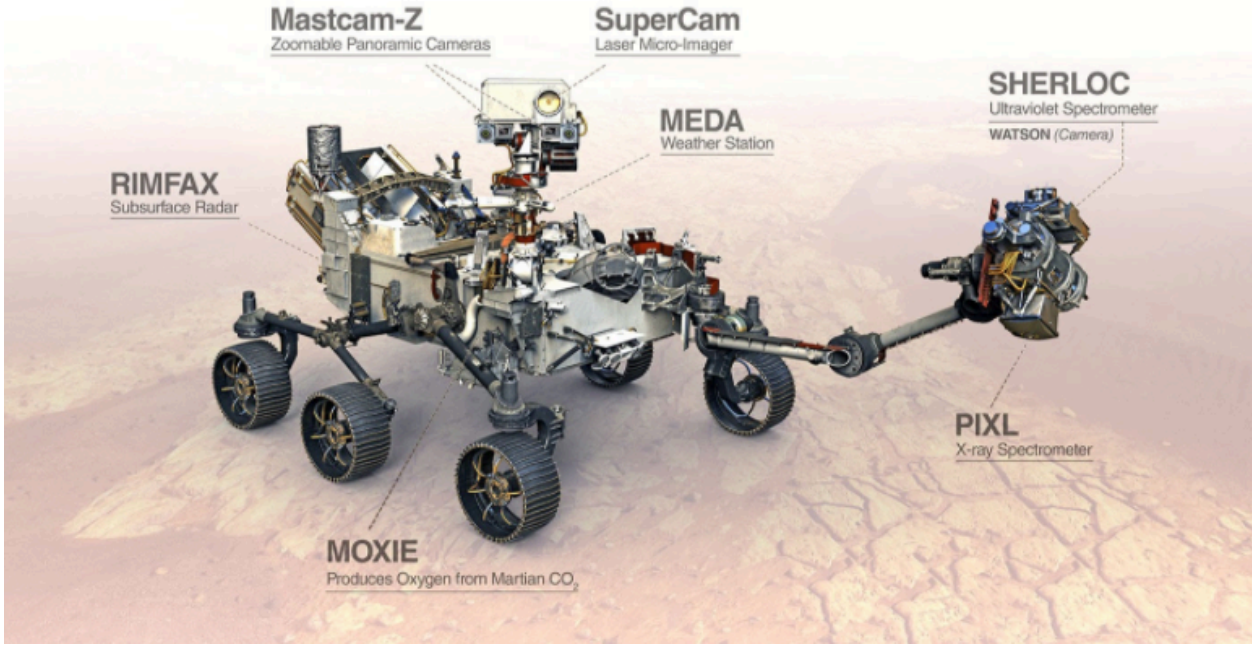


Fig 2: Sensors on a Rover

Difference between Robot and Rover

A rover is a specific type of robot, but the terms aren't interchangeable. A robot is a broad category of any machine that can sense its environment and carry out tasks automatically or semi-automatically. Robots can be stationary, like a robotic arm on a factory assembly line, or mobile, like a delivery robot or a drone.

A rover, on the other hand, is specifically a mobile robot built for locomotion across rugged, uneven, or unpredictable terrain, planetary surfaces, disaster zones, or rough outdoor sites. Its design usually prioritizes mobility and its main purpose is to travel from place to place while carrying out tasks along the way, such as collecting samples or capturing images.

Every rover is a robot, but not every robot is a rover. The distinguishing factor is that a rover's core job is traversal getting across difficult terrain while "robot" covers a much wider range of machines, many of which don't need to move at all.

ASPECT	ROBOT	ROVER
Terrain	Structured or unstructured	Unstructured and outdoor terrain
Mobility	No necessary	Based on its task(always mobile)
Purpose	Manufacturing, automation, entertainment, or assistance	Scientific exploration, soil sampling, and space photography
Power Source	Manufacturing, automation, entertainment, or assistance	Solar panels, radioisotope thermoelectric generators

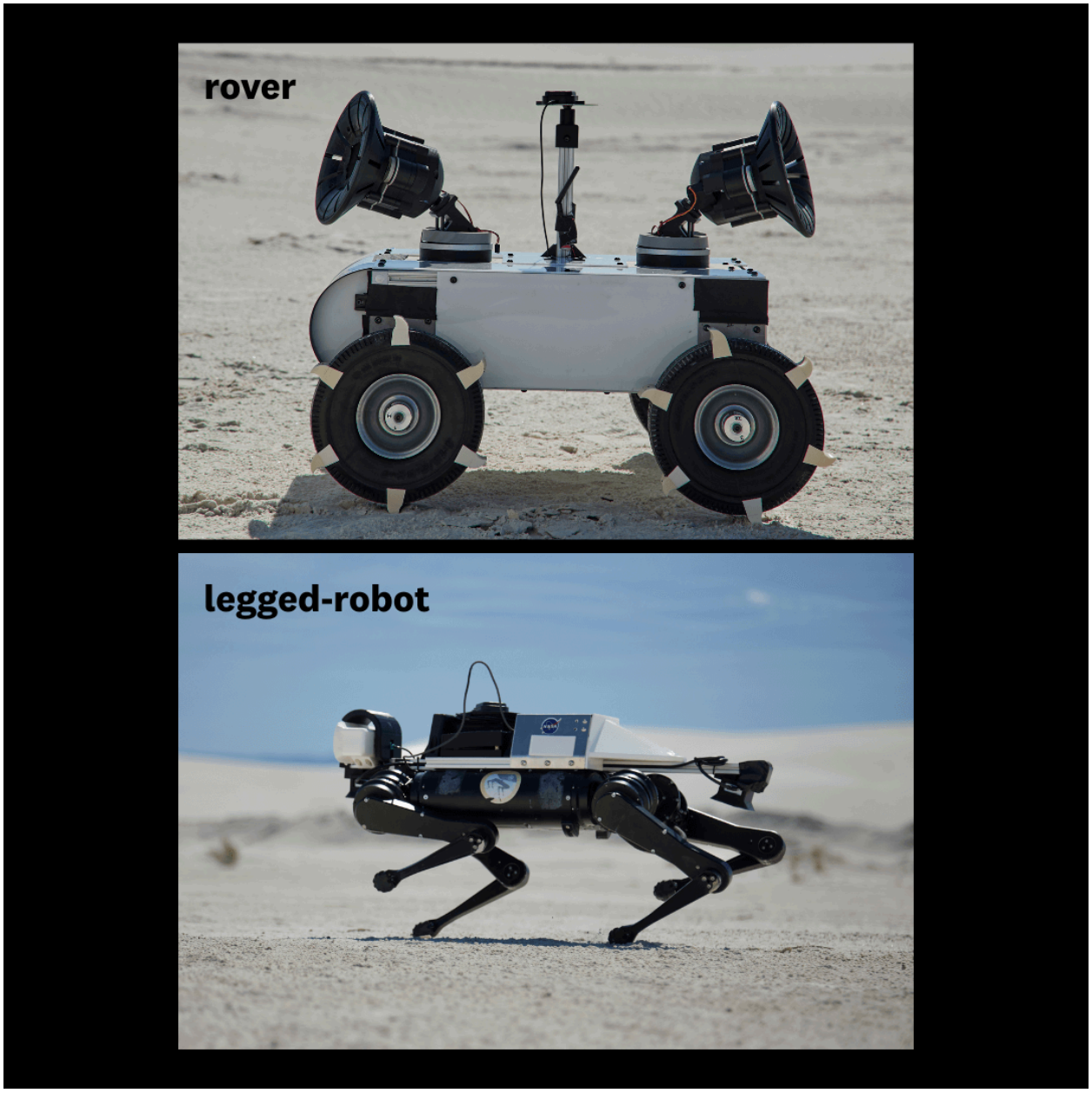


Fig 3: Robot and rover

Autonomous tasks rovers are expected to perform

- Path planning and navigation- Calculating a safe route from one point to another.
- Obstacle detection and avoidance- Reacting to obstacles in real time using sensor data.
- Terrain assessment- Distinguishing safe ground from hazards like slopes, loose rocks, or pits.

- Self-localization- Determining its own position and orientation without relying on external input.
- Object and target recognition- Identifying markers, specific objects, or points of interest.
- Sample collection and manipulation- Using a robotic arm to pick up soil, rocks, or designated items.
- Communication and telemetry- Periodically reporting status and data back to a base station, even through intermittent connectivity.
- Fault detection and recovery- Recognizing when it's stuck and attempting to self-correct.
- Power management- Deciding when to conserve energy or return to a charging point.

Conclusion

Rovers are interesting in that they represent the combination of mechanics, sensors, and autonomy in the form of software. These vehicles are a subset of robots that were designed specifically to be able to cope with the harsh conditions of rough terrain and accomplish important tasks on the go from looking for traces of life on Mars to solving obstacle courses in robotic competitions. With the continuous advancement of sensor technology and processing power, rovers should expect to become more complex autonomous vehicles that dont rely on humans as much.

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